

Probability Using Discrete Random Variables

Statistics Worksheet · Grade 9–12

Name: _____

Date: _____

Learning Objectives

- Identify discrete and continuous random variables in real-world contexts
- Construct a probability model (probability distribution table) for a discrete random variable
- Calculate probabilities using a probability distribution of a discrete random variable

Problems

1. Classify each of the following as a discrete or continuous random variable. (a) The number of students absent in a class on a given day. (b) The height of a randomly selected student. (c) The number of heads when a coin is flipped 5 times. (d) The amount of rainfall in a city in one month.

2. A single fair die is rolled. Let X be the outcome showing on the top face. List all possible values of X and verify that this is a discrete random variable.

$$X \in \{1, 2, 3, 4, 5, 6\}$$

3. A fair coin is flipped once. Let $X = 1$ if the result is Heads and $X = 0$ if the result is Tails. Complete the probability distribution table below.

x	$P(X = x)$
0	
1	

4. A bag contains 3 red marbles and 2 blue marbles. One marble is drawn at random. Let $X = 1$ if the marble is red and $X = 0$ if the marble is blue. Build the probability distribution table and find $P(X = 1)$.

x	$P(X = x)$
0	
1	



5. Two fair coins are tossed simultaneously. Let X be the number of heads obtained. Construct the complete probability distribution table for X .

x	Outcomes	$P(X = x)$
0		
1		
2		

6. The table below shows the probability distribution of X , the number of goals scored by a soccer team in a game. Verify that this is a valid probability distribution, then find $P(X \text{ is at least } 2)$.

x	0	1	2	3	4
$P(X = x)$	0.10	0.25	0.35	0.20	0.10

7. A student randomly guesses on a 3-question true/false quiz. Let X be the number of correct answers. Construct the probability distribution table and find the probability that the student gets exactly 2 correct answers.

$$P(X = k) = \binom{3}{k} \left(\frac{1}{2}\right)^3$$

8. The probability distribution of X is given in the table. Find the missing probability value k , then calculate $P(1 \leq X \leq 3)$.

x	0	1	2	3	4
$P(X = x)$	0.05	0.20	k	0.30	0.15

9. A raffle sells 100 tickets. There is 1 prize of \$500, 2 prizes of \$100, and 5 prizes of \$20. The rest win nothing. Let X be the prize amount won by a randomly selected ticket. Construct the probability distribution and calculate the expected value $E(X)$ — that is, the average amount a ticket holder can expect to win.

$$E(X) = \sum x \cdot P(X = x)$$



10. A discrete random variable X has the probability distribution shown. Determine whether the distribution is valid. If valid, find: (a) $P(X > 2)$, (b) $P(X \text{ is even})$, and (c) the expected value $E(X)$. If not valid, explain why.

x	1	2	3	4	5
$P(X = x)$	0.15	0.25	0.30	0.20	0.10



Probability Using Discrete Random Variables — Answer Key

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Answer Key

1. Answer: (a) Discrete, (b) Continuous, (c) Discrete, (d) Continuous

- A discrete random variable takes countable, separate values (whole numbers).
- A continuous random variable takes any value within an interval (measurable quantities).
- (a) Number of absences — countable whole numbers → Discrete.
- (b) Height — can be any value in a range → Continuous.
- (c) Number of heads — countable outcomes (0–5) → Discrete.
- (d) Rainfall amount — measured on a continuous scale → Continuous.

2. Answer: X can take values 1, 2, 3, 4, 5, 6 — a finite countable set, so X is discrete.

- Rolling a fair die produces one of six outcomes: 1, 2, 3, 4, 5, or 6.
- Each outcome is a distinct whole number — the set is finite and countable.
- Because X can only take specific separated values (not a continuous range), X is a discrete random variable.

3. Answer: $P(X=0) = 1/2$, $P(X=1) = 1/2$

x	$P(X = x)$
0	1/2
1	1/2

- A fair coin has two equally likely outcomes: Heads (H) and Tails (T).
- $P(X = 0) = P(\text{Tails}) = 1/2$.
- $P(X = 1) = P(\text{Heads}) = 1/2$.
- Check: $1/2 + 1/2 = 1$ ✓ — all probabilities sum to 1.

4. Answer: $P(X=0) = 2/5$, $P(X=1) = 3/5$

x	$P(X = x)$
0	2/5
1	3/5

- Total marbles = 3 red + 2 blue = 5.
- $P(X = 1) = P(\text{red}) = 3/5$.
- $P(X = 0) = P(\text{blue}) = 2/5$.
- Verify: $3/5 + 2/5 = 5/5 = 1$ ✓.

5. Answer: $P(X=0)=1/4$, $P(X=1)=2/4=1/2$, $P(X=2)=1/4$



x	Outcomes	$P(X = x)$
0	TT	$1/4$
1	HT, TH	$2/4 = 1/2$
2	HH	$1/4$

- Sample space: {HH, HT, TH, TT} — 4 equally likely outcomes.
- $X = 0$ (no heads): only TT $\rightarrow P = 1/4$.
- $X = 1$ (one head): HT or TH $\rightarrow P = 2/4 = 1/2$.
- $X = 2$ (two heads): only HH $\rightarrow P = 1/4$.
- Sum check: $1/4 + 2/4 + 1/4 = 4/4 = 1 \checkmark$.

6. Answer: Valid distribution. $P(X \geq 2) = 0.65$

- Check validity: $0.10 + 0.25 + 0.35 + 0.20 + 0.10 = 1.00 \checkmark$.
- All probabilities are between 0 and 1 $\checkmark$ — valid distribution.
- $P(X \geq 2) = P(X=2) + P(X=3) + P(X=4) = 0.35 + 0.20 + 0.10 = 0.65$.

7. Answer: $P(X = 2) = 3/8$

- Each question has probability $1/2$ of being correct; questions are independent.
- Using the binomial formula with $n=3$, $p=1/2$:
- $P(X=0) = C(3,0)(1/2)^3 = 1/8$.
- $P(X=1) = C(3,1)(1/2)^3 = 3/8$.
- $P(X=2) = C(3,2)(1/2)^3 = 3/8$.
- $P(X=3) = C(3,3)(1/2)^3 = 1/8$.
- Sum = $1/8 + 3/8 + 3/8 + 1/8 = 8/8 = 1 \checkmark$.
- Answer: $P(X = 2) = 3/8 = 0.375$.

8. Answer: $k = 0.30$; $P(1 \leq X \leq 3) = 0.80$

- The sum of all probabilities must equal 1.
- $0.05 + 0.20 + k + 0.30 + 0.15 = 1$.
- $0.70 + k = 1 \rightarrow k = 0.30$.
- $P(1 \leq X \leq 3) = P(X=1) + P(X=2) + P(X=3) = 0.20 + 0.30 + 0.30 = 0.80$.

9. Answer: $E(X) = \$8.00$

- Identify possible values: $X \in \{0, 20, 100, 500\}$.
- $P(X = 500) = 1/100 = 0.01$.
- $P(X = 100) = 2/100 = 0.02$.
- $P(X = 20) = 5/100 = 0.05$.
- $P(X = 0) = 92/100 = 0.92$.
- Check: $0.01 + 0.02 + 0.05 + 0.92 = 1.00 \checkmark$.
- $E(X) = 500(0.01) + 100(0.02) + 20(0.05) + 0(0.92)$
- $E(X) = 5 + 2 + 1 + 0 = \$8.00$.

10. Answer: Valid. (a) $P(X > 2) = 0.60$, (b) $P(X \text{ even}) = 0.45$, (c) $E(X) = 2.85$

- Check validity: $0.15 + 0.25 + 0.30 + 0.20 + 0.10 = 1.00 \checkmark$. All values in $[0,1] \checkmark$.





- (a) $P(X > 2) = P(X=3) + P(X=4) + P(X=5) = 0.30 + 0.20 + 0.10 = 0.60$.
 - (b) Even values of X: 2 and 4. $P(X \text{ even}) = P(X=2) + P(X=4) = 0.25 + 0.20 = 0.45$.
 - (c) $E(X) = 1(0.15) + 2(0.25) + 3(0.30) + 4(0.20) + 5(0.10)$
 - $E(X) = 0.15 + 0.50 + 0.90 + 0.80 + 0.50 = 2.85$.
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